Git

	Level	Duration
	Introduction	2 days

Course Description

- The main objective of this course is to learn Git, its internals, and how Git works behind the scenes.
- On one hand, Git is the most popular SCM tool but on the other hand, Git is the most complicated one of them all so users need to get familiar with Git and its abilities and internals

Audience and prerequisites

- This course is for anyone who has been using any kind of VCS (Version control system) before.
- `Developers`
- `DevOps`

Main Goals

- How to configure git (configuration, hooks, aliases, gitconfig)
- What are the key features of git (DVCS, 3-states)
- What are branches and how to use them efficiently
- How do merges work (ff, no-ff, rebase)
- git flow and why it is recommended as the daily workflow
- hooks, how to create hooks, and why/when to use them
- git "servers" and clients
- How to improve their productivity and code quality (pull request)
- Tips & Tricks and Beyond

Content

Session	**Content**
Introduction	
	🔹 History
	🔸 The history of git SCSS, RCS
	🔹 Key Features
	🔸 DVCS
	🔸 3-states
	🔸 Storage
	🔸 Heuristics
	🔹 3- states
	🔸 What is 3-states and why do we need it?
	🔸 What can we do with the 3-states
	🔹 Smudge - Clean
	🔹 Configuration
	🔸 Aliases
	🔸 CRLF
	🔸 .gitconfig
	🔸 .gitkeep
	🔸 .gitignore
	Commit
	🔸 What is git commit?
	🔸 How does git store snapshots
Commands/CLI	
	🔹 The basic and most common git commands
	🔸 Init

	🔶 clone
	🔶 add
	🔶 rm
	🔶 commit
	🔶 status
	🔶 checkout
	🔶 log
Branches	
	🔷 What are branches
	🔷 Branches command
	🔶 checkout
	🔶 fetch
	🔶 branch
	🔶 merge
	🔶 pull
	🔶 push
	Hands-on
	Single branch
	Multiple branches
	Single remote branch
	Multiple remote branches
	Multiple branches and multiple users
Merge / Rebase	
	🔷 What is git merge
	🔶 How does git merge branches
	🔶 `ff`

	🔶 `no-ff`
	🔶 `rebase`
	🔶 merge-commit
	🔶 Merge strategies
Git Flow	
	🔷 What is git flow
	🔶 Deep understanding of the git flow model
	🔶 How do we scale it
	🔶 What are the different branches and dots
	🔶 How can we use the gitflow scripts for automating the flow
	🔶 Best practice for git flow
Advanced Topics	
	🔷 assume-unchanged
	🔷 auto-completion / autocorrect
	🔷 bisect
	🔷 cherry-pick
	🔷 clean
	🔷 fake merge / detailed merge
	🔷 git LFS
	🔷 hooks
	🔷 notes
	🔷 reflog
	🔷 rerere
	🔷 squash
	🔷 stash
	🔷 submodule / subtree

	🔹 tags
	🔹 worktree
Pull Request	
	🔹 What is pull request
	🔸 Why should we always use it
	🔸 What does the pull request include
	🔸 CI/CD

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